

The One-Word Census: Answer-Choice Conformity Across 105 Language Models

Tapan Parikh
Cornell Tech
tsp53@cornell.edu

October 2026

Abstract

When a language model must choose one answer from a large space of equally valid options, which answer does it choose, and how often is it the answer every other model chooses? Asked to “pick a word,” 105 language models spanning five years of releases and more than twenty labs chose *serendipity* 46% of the time. We measure this convergence, and each model’s share in it, with a deliberately minimal instrument: 96 single-turn prompts, each naming a category with a wide space of valid one-word answers (“Name a tree.”), asked eight times to each model with no system prompt. The datum is a discrete choice, so analysis is exact match on normalized tokens, with no embeddings and no judge. We score each model by *answer-choice surprisal*: the average $-\log_2$ probability of its answers under the pooled answers of all other models, leave-one-out. In 28 of 96 categories a single answer takes over 80% of all answers across the panel. Conformity still varies widely across models, and the variation is structured. Lightly post-trained and persona-tuned models are the most divergent; heavily post-trained assistants from the major labs are the most conformist, and 27 models give no answer that no other model gave. Within the major providers’ lineages, conformity rises with release order; Anthropic’s generation-5 releases break that trend. Rankings are robust to roster composition (leave-one-family-out $\rho = 0.993$). A greedy-decoding re-run separates stable off-modal defaults from sampling breadth. Models that avoid the modal answer mostly land on the *same* runner-up. After conditioning on a one-dimensional “depth” propensity, no pairwise affinity between any two models survives (0 of 5,460 pairs), while relatives pooled by family share specific rare answers. On open checkpoints of four post-training pipelines, supervised fine-tuning moves each model most of the way to the field’s answers, and later stages change little. Against human category-production norms, the field is more concentrated than people in 18 of 20 shared categories. All prompts, transcripts, and code are public.

1 Introduction

A common complaint about conversational AI is that its answers feel generic, as if they came off a shelf. This paper measures one concrete form of that genericity. When a language model must choose one answer from a large space of equally valid options, which answer does it choose, and how often is that the answer every other model chooses?

The experimental object is deliberately simple. We ask “Name a tree. Reply with one word only.” There are hundreds of familiar trees, and any of them is correct. A population of independent reasoners, or a single reasoner sampling at temperature 1.0, could spread its

probability mass widely. What we observe instead is near-total collapse. Across 105 models from more than twenty labs, *oak* accounts for 95% of all answers. Tools: *hammer*, 95%. Flowers: *rose*, 92%. Vegetables: *carrot*, 92%. With no category constraint at all (“Pick a word.”), the single most common answer, *serendipity*, takes 46% of a pool drawn from the entire English lexicon.

Several studies report that models converge [15, 36, 41], though the strength of the embedding-based evidence is disputed [29]. Our contribution is an instrument with its own null, and what it shows about the structure of the convergence:

1. **A cheap, mechanical, per-model conformity score.** *Answer-choice surprisal* (§3.2) is a leave-one-out measure of how unlikely a model’s answers are under the pooled answers of the rest of the field. It needs no embeddings, no LLM judge and no human annotation. Its noise floor is measured: two halves of the same model’s runs, and two halves of the same field.
2. **A scorecard of 105 models with structured variation** (§4.2). The most divergent models are lightly post-trained or persona-tuned; the most conformist are heavily post-trained assistants from the major labs.
3. **Generational trajectories** (§4.3). Within the major providers’ lineages, later releases are more conformist. Anthropic’s generation-5 releases break the trend.
4. **Robustness** (§4.4, §4.5). Rankings survive leave-one-family-out scoring, roster resampling and a greedy re-run, and a three-level test for pairwise affinity between models comes up empty. Pooled by family, relatives share specific rare answers beyond what their propensities predict.
5. **The runner-up consensus** (§4.5). When models leave the modal answer, they mostly land on the *same* runner-up (*mustard* takes 91% of non-*ketchup* condiment answers).
6. **A human anchor** (§4.6). Against human category-production norms, the field is more concentrated than people in 18 of 20 shared categories.
7. **The favourite word** (§5). Models of every generation name *serendipity* when asked for their favourite word. Newer models volunteer it when asked for any word.
8. **Where in training** (§6). On open stage checkpoints, supervised fine-tuning moves each model to the field’s answers, and the favourite word arrives with post-training and depends on the system prompt.

This is the third version of the census. The first two measured 44 models on 31 prompts at four runs each [25]. This version widens the battery to 96 prompts at eight runs and the panel to 105 models, and adds the training-stage and favourite-word analyses.

2 Related work

The finding: models converge. Jiang et al. [15] report mode collapse on 26K open-ended queries at two levels, intra-model repetition and a stronger inter-model homogeneity, and show that LLM judges and reward models prefer the modal outputs. Schaeffer et al. [29] reexamine

that evidence: under a stricter null, 20 to 32% of pairs of genuinely different answers already cross its embedding-similarity threshold, though a residual effect survives. The census avoids that objection by construction, because it matches one-word answers exactly and reports a within-model and within-field noise floor. Wenger and Kenett [36] find model outputs on creativity tests far more similar to each other than human responses are, and Patel et al. [26] follow the same question across three years of releases with embeddings. Zhang et al. [41] benchmark distributional diversity directly and find larger models within a family often *less* diverse than smaller siblings. Wright et al. [37] find every one of 27 models less epistemically diverse than a web search. Gueorguieva et al. [11] find six models share one discourse-level empathy template. Santurkar et al. [28] show model opinion distributions cluster on a demographic slice, and the CAIS values dashboard [6] finds models rank the world nearly identically in forced-choice elicitation.

Relative to this literature, our instrument trades breadth for three properties. It is *mechanical* (exact match, recomputable from transcripts with no model in the loop), *cheap enough to run at every release*, and *per-model*, producing a scorecard as well as a population-level statement.

Concurrent work fingerprints models with the same kind of prompt. Bruckner [5] collects 30 single-token answers per cell from 165 models on ten trivial prompts in four languages and recovers documented families from the Jensen–Shannon divergence between answer distributions. Output-based phylogenies [39] and text classifiers [33] identify models from what they say. The census scores each answer against the field, so its unit is agreement with other models rather than a model’s identity. Section 4.5 reanalyses Bruckner’s data with the census’s propensity controls.

The mechanism: post-training collapses the mode. Kirk et al. [18] establish that RLHF improves generalisation and reduces output diversity. GX-Chen et al. [13] show that common KL-regularized RLHF settings specify unimodal target distributions. Zhang et al. [40] show instruction-tuned models cannot be random on request. Yang et al. [38] show alignment concentrates probability most at the first positions of a reply, which is all a one-word answer has. Liu [23] localize within-model homogenization to preference optimization, and Karouzos et al. [17] trace where in post-training lineages diversity is lost. Two concurrent papers use the same open stage checkpoints we use. Fortier et al. [10] find convergence present after SFT on the Tülu 3 and OLMo 3 ladders and argue that SFT reveals and amplifies it rather than introducing it. Samanta et al. [27] describe SFT “nucleation” onto a distribution already present in the base model, followed by preference-tuning “settling” within it, on random-generation tasks. Section 6 is a test of that account on answer identity across wide categories. The earliest articulation of mode collapse in RLHF-tuned models is Janus [14]; the generic-response problem predates LLMs [20].

The stakes: convergence propagates. Writers assisted by GPT-4 produce better and more alike stories [9]; LLM-assisted ideation homogenizes without users noticing [1]; the effect traces to the aligned model [24]. Models train on a web increasingly written by models, and recursive training destroys distributional tails first [12, 30]. Diversity-aware post-training now exists [21]. If it lands in production systems, a fixed public instrument run on every release is what would detect the turn.

Methodology. Diversity metrics disagree because “diversity” conflates form and content [34], and our exact-match design responds to that (§3.4). Single-sample evaluation misleads and providers do not reliably honor sampling parameters [31], hence eight samples per cell and separate reporting of conformity and self-consistency. Chen et al. [7] showed that fixed prompt sets detect behavioral drift within months. Transcript-publishing traditions [22, 42, 43] do not hold a fixed instrument over time, and the preservation problem is argued [16] and answered so far only first-party [2].

3 The instrument

3.1 Stimulus

The stimulus is 96 single-turn prompts in two frozen sets. The *census* set is the 31 prompts of v1 and v2: thirty of the form “Name a[n] X. Reply with one word only.” for everyday categories (color, animal, fruit, tree, tool, cheese and so on), and the unconstrained “Pick a word. Reply with one word only.” The *expanded* set adds 65 categories of the same form, including people (painter, scientist, philosopher), places (desert, mountain, lake), abstractions (virtue, vice, emotion) and brands (car brand, brand). There is no system prompt; each prompt is a fresh single-turn conversation; `max_tokens` is 1024 and requested temperature is 1.0. Appendix B lists all 96.

3.2 Answer-choice surprisal

For model m and category c , let $A_{m,c}$ be m ’s answers (up to eight samples) and let the *field* be the multiset of all other models’ answers to c : $F_{-m,c} = \biguplus_{o \neq m} A_{o,c}$. Each answer $a \in A_{m,c}$ is scored by its add-one-smoothed leave-one-out surprisal,

$$s(a) = -\log_2 \frac{\text{count}_{F_{-m,c}}(a) + 1}{|F_{-m,c}| + V_c}, \quad V_c = |\text{support}(F_{-m,c}) \cup \text{support}(A_{m,c})|, \quad (1)$$

and a model’s headline score is the mean of $s(a)$ over all its valid answers across all categories. The score is in bits: each additional bit means the model’s answers are, on average, half as likely under the rest of the field. A model that always gives the field’s answer scores near the floor; a model that reaches for *gouda* where the field says *cheddar* scores higher. Smoothing makes a never-seen answer informative rather than infinitely surprising.

Three companion statistics separate ways of being different:

- **modal avoidance:** the fraction of a model’s answers that differ from the field’s most common answer for that category;
- **novel rate:** the fraction of answers that *no* other model gave;
- **self-distinctness:** distinct answers divided by samples, within model and category.

A model can be original and steady (stable off-modal defaults) or merely noisy (high sampling spread). Self-distinctness is reported rather than corrected away: requested temperature is not honored uniformly across providers [31], so within-model spread doubles as a measured proxy for effective sampling temperature.

Surprisal is relative to the field it is measured in, so each snapshot of the field is pinned by a git tag. Field drift is reported separately (§4.4).

3.3 Normalization and data hygiene

Replies are normalized to a single token: lowercased, stripped of punctuation and emoji, and reduced to the final alphabetic word, which handles models that ignore the one-word instruction (“A common color is blue.” $\rightarrow$ *blue*). A mechanical junk guard treats chat-template artifacts, replies longer than 15 words, bare acknowledgments and single-letter tokens as failed cells. Within each category, bare plurals are merged with their singulars when both occur, and a reviewed table of spelling variants merges forms of one answer (*t-rex* and *tyrannosaurus*). The variant table is new in this version; on the 31 census prompts it changes the model ranking at Spearman 0.999 against v2’s scoring. Of $105 \times 96 \times 8 = 80,640$ attempted cells, 78,348 (97.2%) yield a valid answer.

Uncertainty on each model’s headline score is a bootstrap 90% confidence interval (2000 draws) that resamples categories and pools per-answer surprisals.

3.4 Why one-word answers and exact match

The datum is the *discrete choice* rather than its phrasing. The one-word clamp is identical for every model, so it cannot differentiate them, and it makes the metric immune to verbosity. This avoids the form/content conflation Tevet and Berant [34] warn about. In piloting, an embedding-dispersion metric over free-form answers ranked ernie-4.5 the roster’s clearest content outlier, and inspection showed the distance was carried by verbosity and formatting. Forced to one word, ernie sits in the conformist tail (1.38 bits, 0% novel).

The clamp does not manufacture the convergence. We asked all 137 census, expanded and brand categories without the length instruction (“*Name a tree.*”) on the whole panel and read the first answer each prose reply commits to.¹ Where a strong mode exists, the clamped modal answer leads the free replies about as often as it leads the clamped ones: *oak* 94% clamped and 95% free, *rose* 93%/85%, *apple* 83%/84%, *blue* 81%/77%. Per model, distinctiveness in the free replies rank-correlates with clamped surprisal at Spearman 0.58 ($n = 105$, permutation $p < 0.001$). The exception is categories whose answers are names: asked for one word, the field gives *Picasso* as a painter 59% of the time, while free replies lead with Vincent van Gogh.

3.5 Panel

The panel is 105 models served through OpenRouter, which routes each call to a hosting provider: the model’s own lab for the closed frontier models, or one of several third-party hosts for open weights. It spans the US frontier labs across generations (GPT-3.5 through GPT-6.1; Claude 3 Haiku through Claude Sonnet 5.5; Gemini 2.5 through 3.8; Grok 4.20 through 4.7), Chinese labs (DeepSeek, Qwen, GLM, Kimi, Hunyuan, Ernie, MiniMax, Step, Seed, MiMo, Ling), open-weight families (Llama, Gemma, gpt-oss, Nemotron, Granite, Phi, Mistral, Inkling, Muse), enterprise and search-tuned models (Command A, Palmyra, Sonar), and persona- and community-tuned models (Hermes, WizardLM-2, MythoMax). The roster is append-only. The first 44 models’ census runs 1–4 date from July 2026; every other cell was collected between September and early October 2026. Each transcript is stamped with its date. The full list with metadata is in the repository.

¹Script: `probe_clamp.py`; `probes/clamp_rank.json`.

4 Results

4.1 The substrate: a strong mode almost everywhere

Figure 1 shows how concentrated each category’s pooled answers are. In 28 of 96 categories a single answer accounts for at least 80% of every answer given by every model: *Sahara* (99%), *quartz* (98%), *Orion* (97%), *Everest* (96%), *Zeus* (96%), *oak* (95%), *hammer* (95%), *granite* (94%), *Toyota* (94%), *basil* (93%), *rose* (92%), *carrot* (92%), and sixteen more. The median category’s modal answer takes 58%. At the diffuse end are categories with no single prototype: virtue (*honesty*, 23%), reptile (27%), sound (29%), gemstone (30%), smell (30%), country (31%).

The unconstrained prompt is the sharpest case. Given the entire English lexicon, the field still collapses: *serendipity* takes 372 of 816 answers (46%), followed distantly by *apple* (44), *ephemeral* (30), *hello* (25) and *lantern* (24), with 122 distinct words in total.

The mode is the frictionless answer. The modal answer is not the most common referent in the world, nor the most discussed. It is the blandest *unambiguous* exemplar. Fruit: *apple* 685, *mango* 110, *banana* 27, *orange* 3 of 828. Orange, among the most common fruits in life and text, nearly vanishes, plausibly because it collides with the color. Vegetable: *carrot* 759, *broccoli* 57, *tomato* 2 of 828.² In a panel dominated by US-lab models, the United States is named twice in 820 country answers; the field’s answers are *Japan* 250, *France* 231 and *Canada* 231. Answers with an edge, such as a homonym, a taxonomic argument or a connotation, are systematically absent.

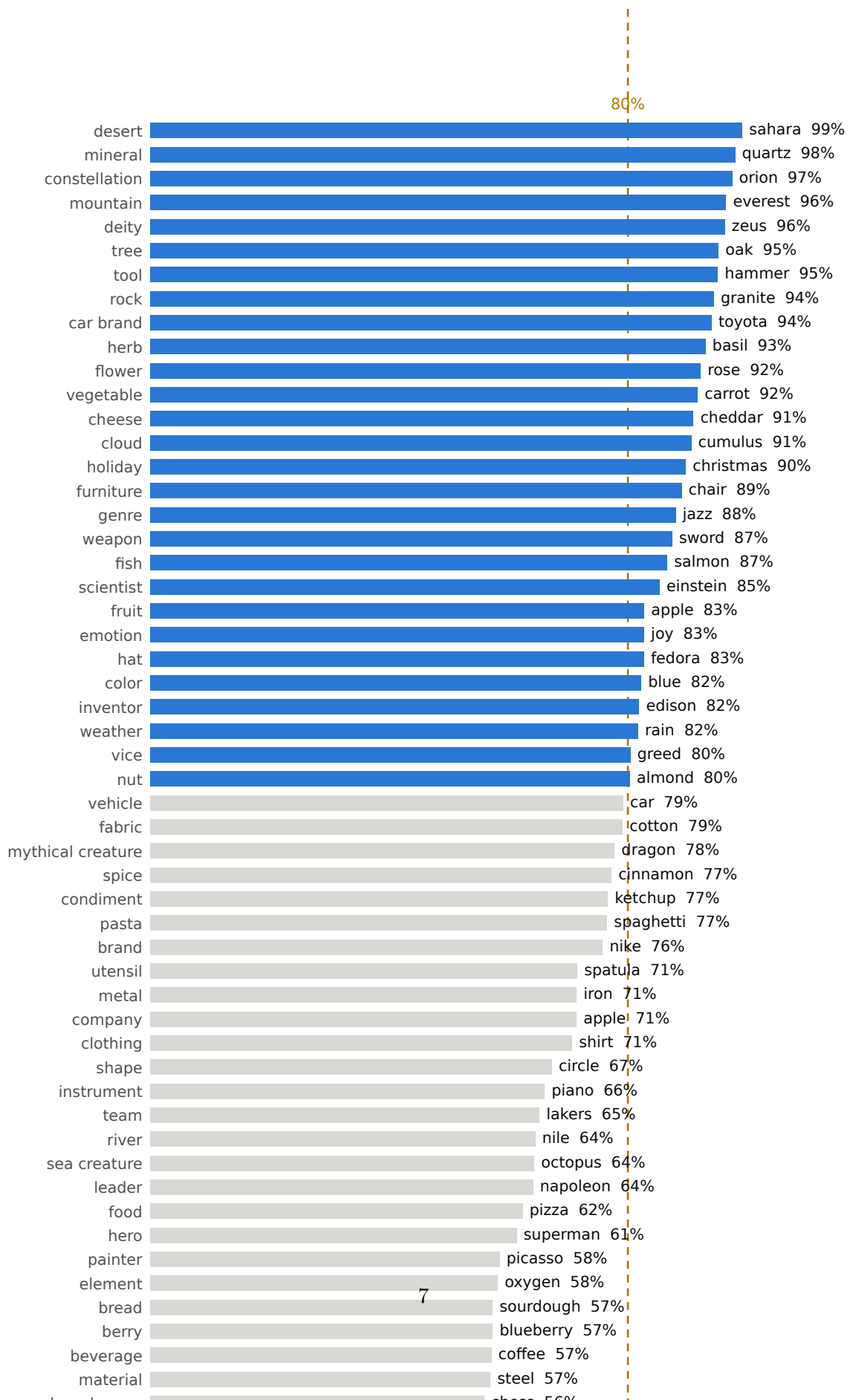
Nor is the mode corpus frequency’s pick. A deflationary null says models echo word frequency. Scored against a multi-corpus frequency lexicon [32], the null fails on the field’s own answers: the modal answer is the most corpus-frequent member of the field’s support in only 14 of 96 categories.³ *Potato* (Zipf 4.11) outranks *carrot* (3.62) and is 4 of 828 vegetable answers, and *serendipity* ranks 113th of the 122 answers to the unconstrained prompt by corpus frequency. Frequency does shape the field’s *support*: within categories, pooled answer counts correlate with Zipf frequency (mean Spearman +0.30, positive in 82 of 95 categories). It neither selects the winner nor explains the winner’s dominance. Across categories, modal share is uncorrelated with the modal word’s frequency ($\rho = +0.08$), and a sampler drawing answers in proportion to corpus frequency over the same support would give the field’s modal answer a mean share of 12%, against the observed 63%. Both biases in this test favour the null: the candidate set is restricted to answers the field produced, and raw unigram frequency credits homographs. Frequency is a background availability prior; the selection among available answers is made by something else.

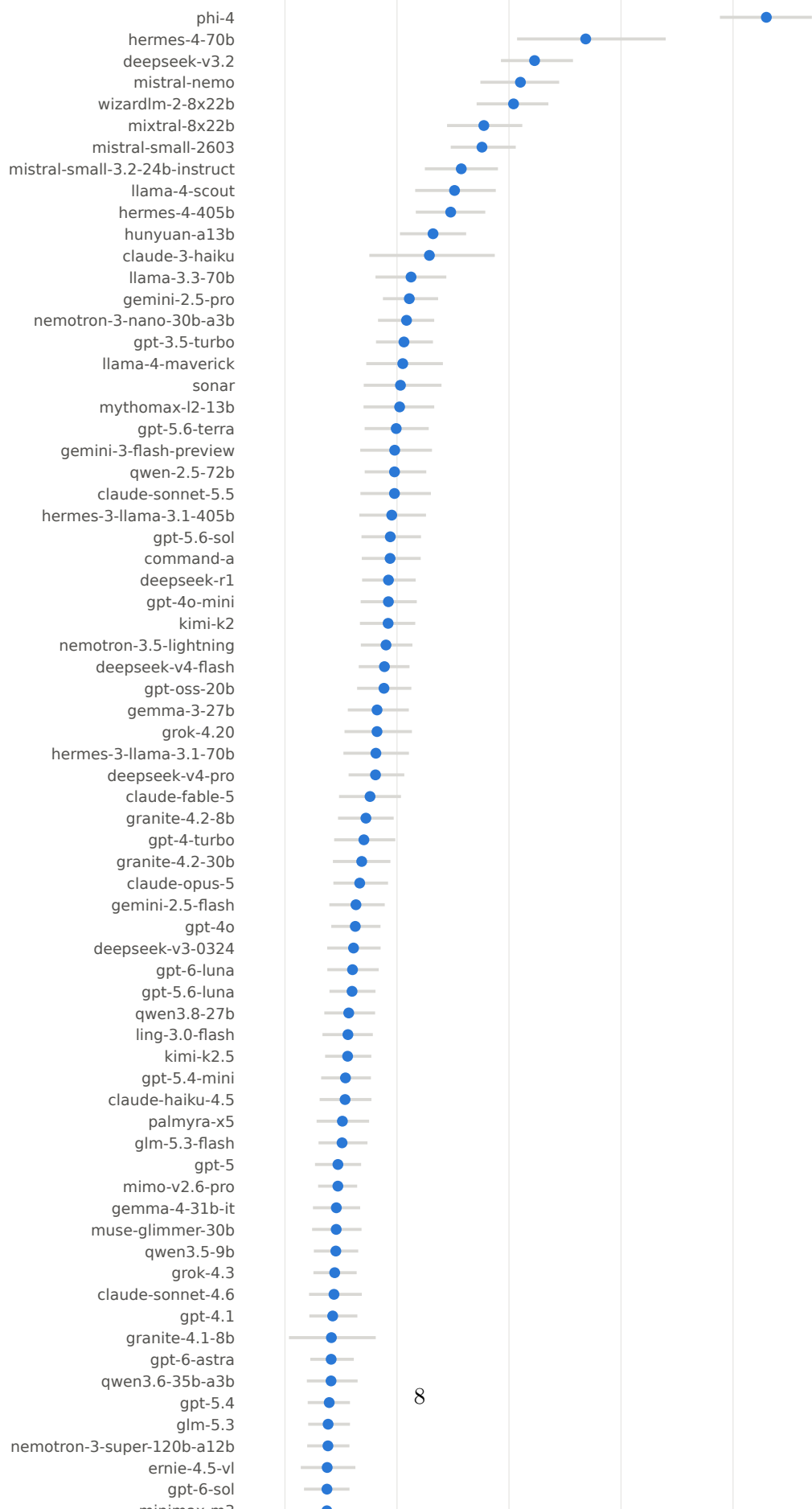
4.2 The scorecard: conformity varies widely, and the variation is structured

Figure 2 and Table 3 (appendix) give the full scorecard. Scores run from 1.00 to 5.30 bits. Without the one outlier at the top, the range is 1.00 to 3.68 bits, a factor of about six in answer likelihood under the field.

²*Tomato* is botanically a fruit, so a model may be right to omit it. *Orange* carries the point without that confound.

³Script: `probe_corpusfreq.py -v3`. Candidates for each category are the field’s support (every distinct answer any model gave), so no external category list is needed.





The divergent tail is lightly tuned and persona-tuned. Phi-4 heads the scorecard at 5.30 bits, with 32% of its answers given by no other model. Many are coinages (*chaoswright* for a villain) or repeat the category word (*hat* for a hat), and at greedy decoding it loses 2 bits (§4.4). Below it are Hermes 4 70B (3.68 bits; 16% novel; 61% modal avoidance), DeepSeek v3.2 (3.23), Mistral Nemo (3.10), WizardLM-2 (3.04) and Mixtral 8x22B (2.78). By panel origin, persona- and community-tuned models average 2.50 bits and small open models 2.33, against 1.54 for US frontier models, 1.56 for enterprise models and 1.59 for Chinese-lab models. The one search-grounded model, Sonar, scores 2.03. Most of the divergent tail is *distributional*: re-scored at greedy decoding, these models fall back toward the field (§4.4). In the terms of §3.2 they diverge in what they sample more than in what they say first.

The conformist tail is heavily post-trained assistants. The bottom of the scorecard is Claude Opus 4.5 (1.00), Opus 4.7 (1.12), Gemini 3.6 Flash (1.13), Opus 4.6 (1.14), Muse Spark 1.3 (1.14) and Grok 4.5 (1.17). These are mainline assistants from the major labs, and 27 models in the panel give no answer that no other model gave. Opus 4.5 avoids the modal answer on 21% of its answers.

Exceptions within a lab. Two same-generation pairs separate. Claude Fable 5 scores 1.76 against Claude Sonnet 5’s 1.19 (paired permutation over categories, 0.57 bits, $p < 0.001$).⁴ Its divergence is stable rather than sampled: *gouda* for cheese and *mango* for fruit in eight runs of eight, with self-distinctness of 0.20. The GPT-5.6 tiers released the same day also spread: Luna 1.60, Sol 1.94, Terra 1.99 (Sol against Luna, 0.34 bits, $p < 0.001$). Terra shares some of Fable’s stable defaults (*mango* eight of eight). Neither ordering holds across generations: Claude Sonnet 5.5 (1.98) is more divergent than Fable 5.1 (1.30), and the GPT-6 tiers sit between 1.37 and 1.60. The two pairs are exceptions the instrument detects and does not explain.

4.3 Generational trajectories

Figure 3 tracks surprisal across releases within each lineage. Pooling across families could manufacture a generational trend if newer release windows simply hold more conformist labs, so we test within lineages. Over the six major providers (Claude, GPT, Gemini, Grok, Qwen and DeepSeek; 61 models), surprisal falls with release order: the family-demeaned Spearman correlation is -0.30 (within-family permutation null, 2,000 draws, $p = 0.025$).⁵ The decline is steepest in Gemini (-0.88 ; 2.5 Pro 2.11 to 3.6 Flash 1.13), Grok (-0.60 ; 4.20 1.82 to 4.5 1.17, then flat) and GPT (-0.47 ; GPT-3.5 2.06 to GPT-6.1 Sol 1.24). Qwen falls from 1.98 (2.5-72B) to between 1.20 and 1.57 for the Qwen 3 releases. DeepSeek is non-monotonic.

Claude follows the trend and then leaves it. Its lineage falls from Claude 3 Haiku (2.29) through the Opus 4.x releases to the panel’s floor (Opus 4.5 1.00, Opus 4.7 1.12, Opus 4.8 1.21; Spearman -0.53 over its first nine releases). The generation-5 releases then move back off the consensus: Fable 5 1.76, Opus 5 1.67, Sonnet 5.5 1.98. With Claude’s walk cut before Fable 5, the major-provider correlation is -0.46 ($p < 0.001$), and without Claude it is -0.47 ($p = 0.002$). The within-lab exceptions of §4.2 sit inside this break.

Where the decline occurs it is sharpest where a strong mode exists. On the most concentrated categories (modal share $\geq 80\%$), each lineage’s newest member averages 0.51 bits against 0.92

⁴Script: `probe_permutation.py -v3`. Shuffling the two models’ labels within each category, 10,000 draws.

⁵Script: `lineage_trend.py`. Across all 13 families with three or more models (85 models) the correlation is -0.20 ($p = 0.074$); the smaller open families (GLM, Granite, Hermes) run the other way.

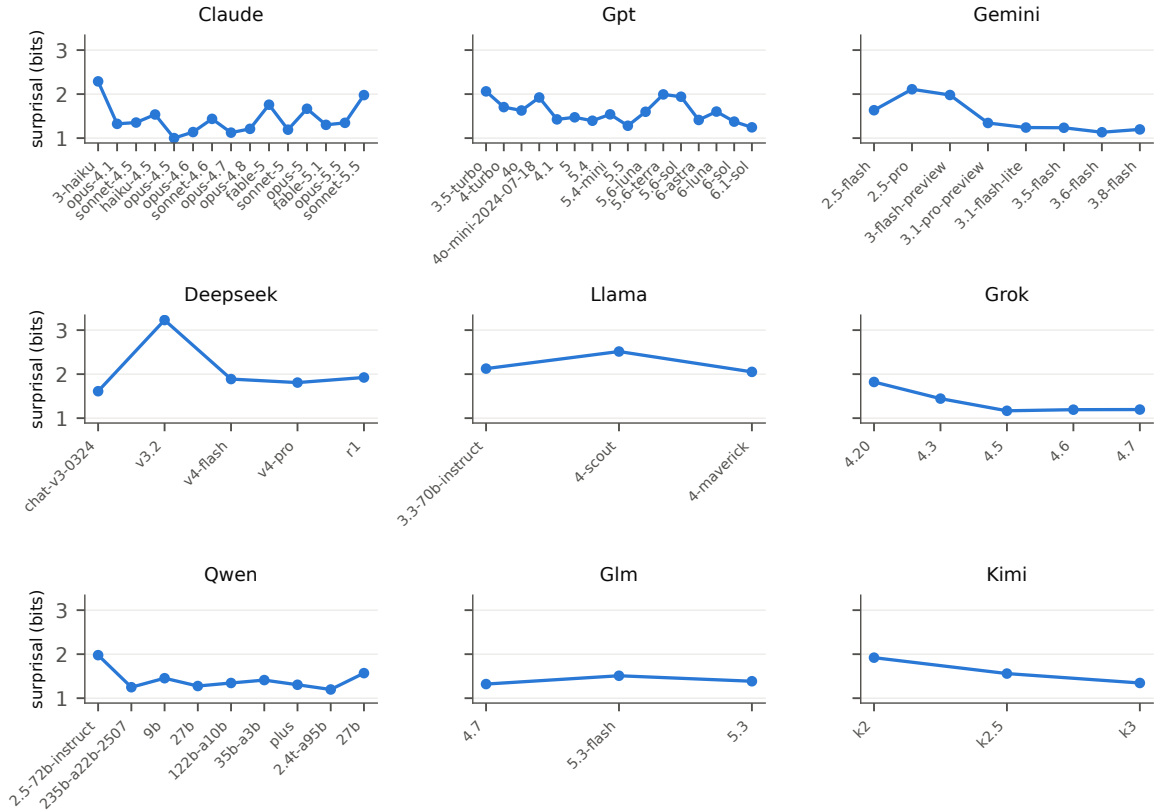


Figure 3: Answer-choice surprisal across releases within each tracked lineage, in release order (shared y-axis).

for its oldest; on the diffuse categories ($\leq 40\%$) the same cohorts score 2.57 against 2.83. The direction is consistent with the bigger-is-less-diverse finding of Zhang et al. [41] and with the post-training literature (§2). The roster does not separate size from generation, and each model is measured at one date.

Within one lineage, post-training moves the score. The DeepSeek lineage is the caveat against reading these walks as family destiny. v3-0324 scores 1.61, v3.2 jumps to 3.23 (the panel’s third most divergent model), R1 sits at 1.92, and v4-flash and v4-pro return to 1.89 and 1.81. At the same requested temperature, v3-0324 gives eight identical answers in 35 of 96 categories, while v3.2 does so in 7 and scatters into the tail (*Esperanto* and *Swahili* for a language, *Charleston* for a dance, *crimson* for a color). Their modal answers differ in 40 of 96 categories. At greedy decoding the gap narrows (v3.2 2.03, v3-0324 1.48; §4.4): the v3.2 recipe mostly widened the sampled distribution.⁶ Per DeepSeek’s release documentation [8], v3-0324 is a post-training refresh of the V3 base, and v3.2 descends from it through continued training and a new RL recipe, with no separate pretraining run. The gap between a conformist and one of the panel’s most divergent models is attributable to training after the base.

⁶In v2 we re-ran a subset of categories with both models pinned to the same host; the contrast reproduced, so it is not an artifact of which provider served each call.

4.4 Robustness

Because scoring is leave-one-out against the pooled panel, a fair objection is that the scorecard measures the roster. Absolute bit values are relative to this field by construction. We test whether the *rankings* are roster artifacts, with no new API calls (`robustness.py -v3`):

- **Leave-one-family-out.** Rescoring every model against a field with all its relatives removed yields Spearman $\rho = 0.993$ against the shipped ranking. Sibling contamination is small and concentrated in GPT, the largest family (16 models): GPT-5.6 Sol gains 0.16 bits and moves from rank 25 to 17. Phi-4 is first and Opus 4.5 last either way.
- **Balanced fields.** Against one randomly drawn model per family (200 draws), the top three rank 1–3 in every draw and Opus 4.5 ranks last in every draw.
- **Random subsets.** Against random 15-model fields (200 draws), the top three hold and Opus 4.5’s median rank is 105 (5th–95th percentile 102–105).
- **Era-balanced fields.** Against fields with equal numbers of models from three release eras, which strips the numerous newest models of their weight in the mode, the ranking holds at $\rho = 0.986$ and Opus 4.5 is last (104–105).
- **Split halves.** Scoring each model on half its categories against the other half gives a median Spearman of 0.87 (0.93 after the Spearman–Brown correction).
- **Smoothing.** Against add-0.5, add-2 and add-0.1 smoothing and an alternative V_c , $\rho \geq 0.998$, with the same top six and last model (`probe_smoothing.py -v3`).

The panel is an availability sample of what the aggregator serves, without usage weighting. Usage weighting would make the reference field more dominated by the major labs’ assistants and sharpen the contrasts reported here.

Sampling spread and the greedy re-run. Mean surprisal is coupled to within-model spread: a model that samples more widely hits lower-probability answers more often, and across the panel self-distinctness and surprisal correlate at $r = 0.85$ (Spearman 0.70). Since requested temperature is not honored uniformly, part of the scorecard could be effective serving temperature. We re-ran the full battery at requested temperature 0 and scored each model’s greedy answers against the frozen temperature-1 field.⁷ The re-run is also a temperature-compliance census: 58 of 105 routes honor it. The other 47, including every Claude model and every GPT model from GPT-5 on, return unchanged spread, so for them the re-run is a replication rather than a control. The persona-tuned and lightly tuned models collapse toward the field: WizardLM-2 by 2.11 bits (to 0.94, last at greedy), Phi-4 by 1.98 (to 3.32, still first), Hermes 4 70B by 1.85 and DeepSeek v3.2 by 1.20. Their divergence is real in what they sample and largely absent from their greedy defaults. The two scorecards correlate at $\rho = 0.79$. Opus 4.5 stays at the floor (0.99), and the Fable 5–Sonnet 5 gap persists (+0.47 bits). Conformity at the bottom cannot be a temperature artifact, since nothing is more conformist at temperature 1 than it is at greedy. Divergence at the top comes in two kinds the instrument separates: stable off-modal defaults (Fable 5, the GPT-5.6 tiers, the Llamas) and sampling breadth (the persona tier), which is the first thing a greedy serving configuration removes.

⁷Script: `probe_temp0.py -v3`; transcripts in `transcripts-temp0/` and `transcripts-expanded-temp0/`. “Honored” means self-distinctness over four runs drops by at least 0.05.

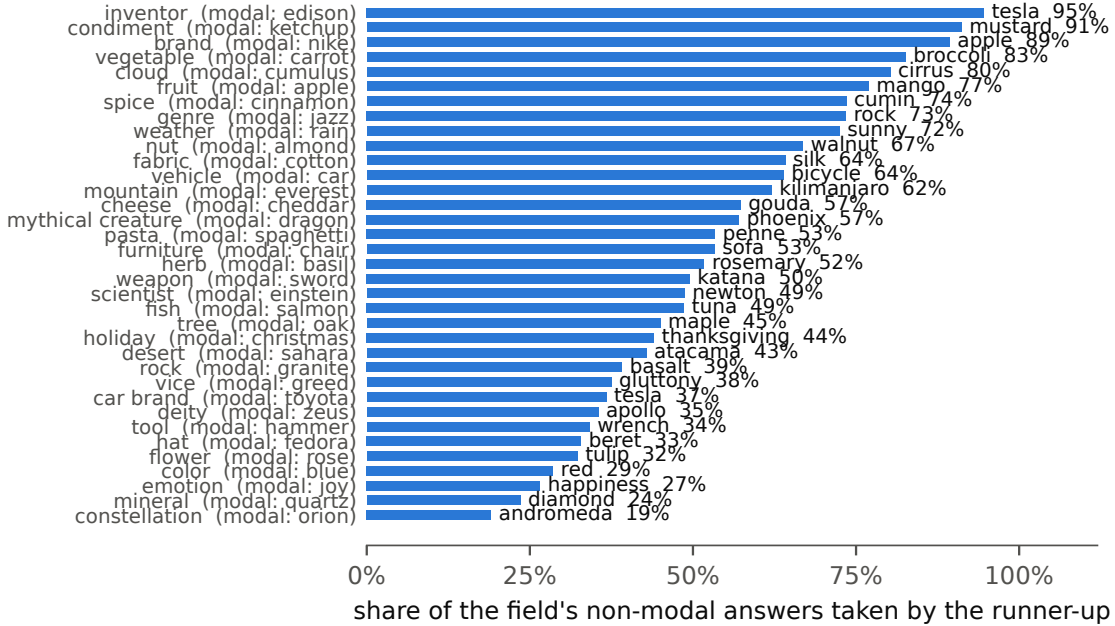


Figure 4: The runner-up consensus. For each high-concentration category (modal share $\geq 75\%$), the share of the field’s *non-modal* answers taken by the single most common non-modal answer.

Field drift. Surprisal is field-relative, so a change in a model’s score can come from the model or from the field. We measured the field’s own movement on the v2 models still served (41 of 44): their July runs against their September–October runs on the 31 census prompts. The mean Jensen–Shannon divergence between the two answer distributions is 0.031 bits per category, against 0.028 between two random halves of the same runs. The modal answer changed in 2 of 31 categories (*gemstone*, *sapphire* to *diamond*; *insect*, *ant* to *butterfly*). Over three months the field’s answers moved no more than resampling moves them. Across the 44 shared models, the v2 and v3 rankings correlate at Spearman 0.89, though v3 adds 65 categories and four runs.

4.5 Pairwise affinity, family resemblance, and the runner-up consensus

We tested for pairwise affinity across all $\binom{105}{2} = 5,460$ model pairs with a three-level ladder, each level conditioning away a population structure the previous level mistook for affinity (Benjamini–Hochberg FDR, $q = .05$; Monte Carlo nulls where normal approximations fail on small counts; `pairwise.py -v3`). Level 1 (rarity-weighted co-occurrence against a random-partner null) leaves 259 apparent pairs, but is confounded by avoidance propensity: two habitual modal-avoiders co-occur against a conformist null with no coupling at all. Level 2 (conditioning on avoidance) leaves 515, led by same-family pairs (Hermes 3 405B $\times$ Hermes 4 405B, GPT-5.6 Sol $\times$ GPT-6 Sol, Llama 3.3 $\times$ Llama 4), but is confounded by a *depth* propensity: how far down the field’s answer distribution a model lands when off-modal is a stable per-model trait. Level 3, conditioning on both: **zero of 5,460 pairs survive.**

The conditioning variables are themselves findings:

1. **The runner-up consensus** (Figure 4). The off-modal distribution has its own mode:

Tesla takes 95% of non-*Edison* inventor answers, *mustard* 91% of non-*ketchup*, *Apple* 89% of non-*Nike* brand answers, *broccoli* 83% of non-*carrot*, *mango* 77% of non-*apple*.

2. **Depth propensity is one-dimensional.** Among each model’s distinct off-modal answers, the fraction landing on the field’s #2–#3 answers is a stable trait: Opus 4.5 82%, Grok 4.5 80%, Opus 4.8 77%, Sonnet 5 72%, Fable 5 65%, against Hermes 4 70B 36%, WizardLM-2 36%, DeepSeek v3.2 33% and Phi-4 18%.

Family resemblance survives the propensities. Pooling pairs by family recovers structure that single pairs do not. Within-family pairs share rarity-weighted off-modal answers beyond between-family pairs by $D = 6.01$, against 2.66 ± 0.04 under a null that keeps each answer’s field-rank band (modal, #2–#3, deeper) and redraws the word within it ($p < 0.001$, 1,000 draws).⁸ A model’s nearest neighbour is a relative for 56 of 93 models in families of two or more, against 43.0 ± 2.2 under the same null. Relatives share particular rare answers beyond what their avoidance and depth propensities predict.

Bruckner’s released data [4] allow the same separation at a sampling depth the census lacks. Rerunning his Jensen–Shannon family recovery reproduces his figure (97 of 162, 59.9%).⁹ A null that keeps only each model’s modal, runner-up and deep-tail rates in every cell, and redraws the words, already recovers 88 of those 97. His fingerprint is therefore mostly the two propensities described above. Restricting similarity to shared off-modal answers, weighted by rarity, recovers 98 of 162 families, and the same null explains only 52 of them. At 30 samples per cell, specific shared answers carry about half of the family signal.

Prompting reaches conditional modes. Asking for divergence opens a new column with its own mode. Prompted “*Name an unusual fruit,*” the 44-model v2 panel produced 12 distinct fruits: *durian* takes 44%, *rambutan* 34%, and the top three take 84%. Conditioning on unusualness triples the distinct-answer count of the plain fruit prompt, and the mass re-concentrates on the prototypes of the new frame. This fits the finding of Zhang et al. [40] that the collapse lives in the weights and prompting navigates between conditional modes.

4.6 Compared to a human population

The Battig–Montague category-production norms [3], updated by Van Overschelde et al. [35], record for each exemplar the proportion of participants who produced it as their *first* response, which gives a single-response human distribution for 20 of our categories. For the six whose norm wording differs from ours (“a four-footed animal,” “a carpenter’s tool,” “a precious stone,” and three more), we use an exact-wording rerun.¹⁰

The model field is more concentrated than people in **18 of 20** categories (Figure 5). The modal answer takes 69% of the field on average against 36% of human first responses, and people carry more distinct answers above 5% (4.8 against 2.8). The gap is widest where a strong textual prototype exists: *oak* is 95% of the model field and 31% of first human responses. One reversal is marginal (*occupation: doctor* 33% of people, 32% of models). The other is *country*,

⁸Script: `family_signal.py -v3`. On the v2 panel the same script reproduces v2’s $D = 1.15$.

⁹With his released family labels, except that `writer/palmyra-x5`, which a name pattern had assigned to the Gemini family, is treated as its own family.

¹⁰The wording matters. Our wording “Name a gemstone” gives one of the flattest modes in the battery (*diamond* 30%, *sapphire* 27%, *ruby* 21%, *emerald* 14%), while the norm wording “a precious stone” gives *diamond* 88%.

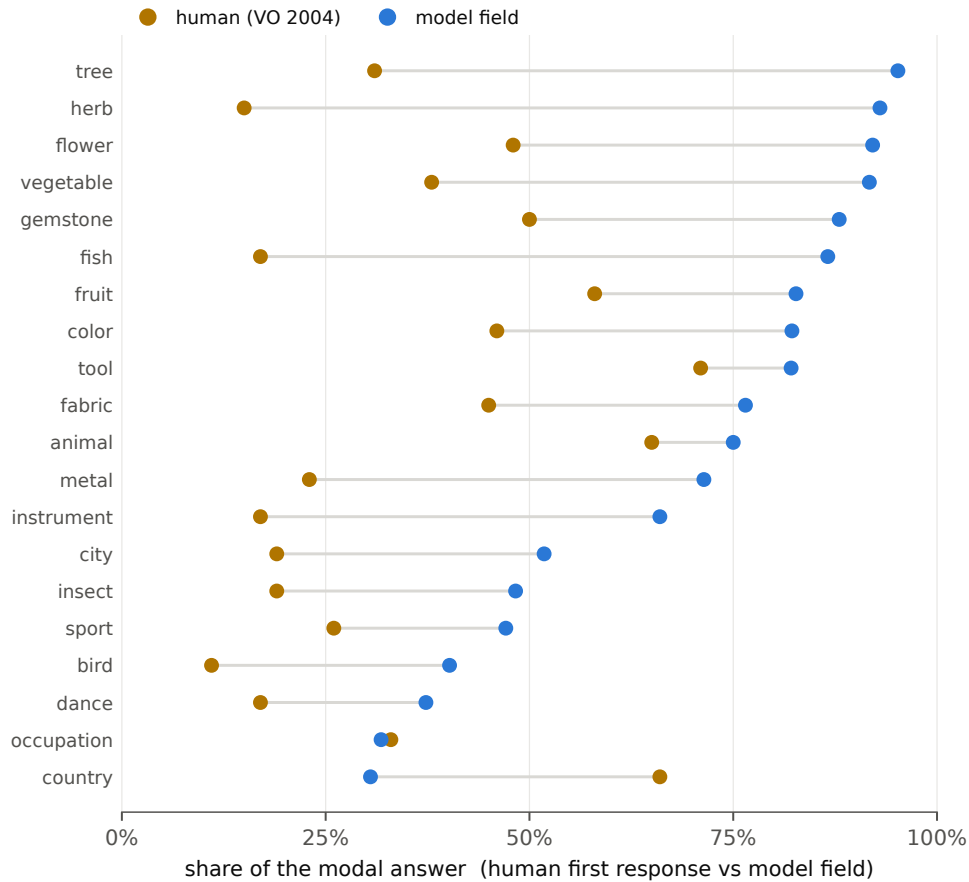


Figure 5: Concentration of the modal answer, model field against a human population, on the 20 categories shared with the Van Overschelde norms. Human = the fraction of people who gave the answer as their *first* response; model = modal share of the 105-model field (norm wording for the six categories whose wording differs).

the frictionless mode of §4.1 seen from the other side: 66% of people name their own country, while the models spread across *Japan*, *France* and *Canada*. *Tomato* is an 11% first response for “a vegetable” among people and 0.2% of the model field.

Pooling a model’s eight runs with every other model’s mixes within-model and between-model spread, while each person contributes one answer. With one vote per model (its most frequent answer) on the 14 categories whose full human distribution is available, the field’s top answer takes 75% of model votes against 34% of people for theirs, the two populations share a top answer in 5 of 14 categories, and the effective number of answers is 2.2 for models against 8.9 for people. Models choose among human answers: 92% of model votes name an answer some person also gave.¹¹ The norms are US undergraduates around 2004, a demographically narrow population, so a broader one would likely spread its answers further.

¹¹Script: `analyze_humannorms.py -v3`. The norms are copyrighted and not redistributed; the repository holds derived per-category numbers only.

Table 1: Share of replies containing *serendipity*, by prompt and release date (`probe_serendipity.py`). Each model’s share is averaged within the cohort. “Say any word” was asked of 83 models.

prompt	all	before 2025 (14)	2025 (28)	2026 H1 (32)	2026 H2 (31)
“Say any word.”	20%	5%	14%	19%	35%
“Pick any word.”	41%	20%	37%	52%	43%
“Pick a word.” (census)	44%	24%	42%	52%	48%
“What is your favorite word?”	65%	61%	57%	75%	64%
“Most beautiful word in English?”	72%	59%	68%	73%	79%

5 The favourite word

Serendipity is the sharpest single consensus in the census, and it has a history. Table 1 asks for a word in five ways, from neutral to explicitly evaluative, and splits the panel by release date. Asked directly for a favourite, models of every era name *serendipity* (61% of models released before 2025, 64% of those released in the second half of 2026). The neutral prompts rise across generations: “Pick any word” from 20% to 52% by the first half of 2026, and “Say any word” from 5% to 35%. GPT-3.5 names it as its favourite word in seven of eight replies and offers it unasked in one of eight. GPT-6.1 Sol gives it in six to eight of eight for every prompt. Older models know the favourite; newer ones volunteer it when asked for any word.

The rise holds within lineages. Over the six major providers, the share for “Pick any word” rises with release order (family-demeaned Spearman $+0.36$, $p = 0.012$; the census wording gives $+0.32$, $p = 0.023$). Anthropic’s newest releases are the exception, as in §4.3: Claude Opus 5.5 and Sonnet 5.5 never give *serendipity* for “Pick a word,” and *lantern*, the fifth most common answer in the field, comes mostly from Claude.

Across languages (exploratory). The same “pick any word” prompt was asked in 44 languages, translated by a model and not yet checked by native speakers. Translated forms of *serendipity* appear in Portuguese (10% of replies across the panel), Spanish (7%), Italian (4%) and French (2%), more often from 2026 releases (Portuguese 13% and Spanish 11% in the first half of 2026, against 6% and 1% before 2025). German stays at 1–4%. In most other languages the field settles on a plain pleasant word (*peace, love, light, sky*). English is the only language whose favourite is rare in ordinary text.

6 Where in training

The census is black-box, but open releases that publish a checkpoint after each post-training stage let us place the convergence. We sampled four pipelines locally: OLMo 3.1 32B and OLMo 3 7B (base, SFT, DPO, RL), Tülu 3 8B (base, SFT, DPO, RL), and Nemotron 3.5 Lightning (base and final). Each stage answered all 96 prompts 20 times and was scored against the 105-model API field as a model outside the panel.

SFT does most of the work. Table 2 shows the same shape in every pipeline. The largest step toward the field is supervised fine-tuning: OLMo 3.1 32B goes from 5.88 bits at base to 2.44 at SFT, and its share on the field’s modal answer from 0.23 to 0.54. DPO and RL change

Table 2: Open training-stage checkpoints scored against the 105-model field on the 96 prompts, 20 samples per prompt, no system prompt (`stage_ladder.py census`). Surprisal is mean bits against the whole field; modal share is the share of answers matching the field’s modal answer; entropy is the stage’s own answer entropy per category. Base models are sampled as raw completions, which inflates their surprisal.

pipeline	stage	surprisal	modal share	entropy
OLMo 3.1 32B	base	5.88	0.23	3.19
	SFT	2.44	0.54	1.35
	DPO	2.18	0.58	1.15
	RL	2.15	0.57	1.09
OLMo 3 7B	base	5.82	0.25	3.04
	SFT	3.74	0.37	1.87
	DPO	3.36	0.43	1.49
	RL	3.26	0.42	1.43
Tülu 3 8B	base	8.69	0.05	3.92
	SFT	5.62	0.22	2.93
	DPO	5.75	0.19	2.88
	RL	5.92	0.18	3.00
Nemotron 3.5 Lightning	base	6.81	0.15	3.41
	final	3.57	0.40	2.04

little after that (2.18, 2.15), and in Tülu they move slightly back off the field. Each stage’s own entropy falls in step with its distance from the field, so conformity and diversity loss travel together. No open checkpoint reaches the field’s median (1.51 bits); the closest is the final OLMo 3.1 32B at 2.15. This matches the account of Fortier et al. [10] and Samanta et al. [27]: the base already ranks the field’s answers high, and SFT fixes the model on them. The base models do lean toward the field’s answers (OLMo base gives the field’s modal answer about a quarter of the time), which is consistent with SFT selecting a distribution already present.

The favourite word arrives with post-training and depends on the frame. Asked “Pick a word” 200 times per stage, the OLMo 3.1 32B base never says *serendipity* (0 of 199). SFT installs it as the top answer under every system prompt we tried (14–22%). After that, the share depends on the system prompt more than on the stage. With the system prompt that every post-training example carried (“You are a helpful function-calling AI assistant...”), it is 18% at SFT, 22% at DPO and 9% after RL. With the identity prompt the released model ships with (“You are Olmo...”), which no training example carried, it is 22%, 20% and 30%. With no system prompt it falls from 14% to 2%. On the same weights, the released template gives the word three times as often as the training template.¹² Asked for its favourite word under the training prompt, the same ladder gives *serendipity* 1% at base, 30% at SFT, 52% at DPO and 45% after RL. The smaller pipelines barely volunteer it: OLMo 3 7B gives it 2, 1 and 0 times in 50 at SFT, DPO and RL, and Tülu never does. Nemotron 3.5 Lightning names it as the most beautiful English word 23 times in 50 and gives it once in 50 for “Pick a word.” Like the older API models in Table 1, these models know the favourite and do not volunteer it.

¹²Scripts: `probe_olmo_ladder.py`; `probes/olmo_ladder_32b_n200.json`. The training system prompt was read from the Dolci data and open-instruct’s chat template.

Evaluations of post-trained models should therefore report and match the deployed chat template and system prompt. The same weights give 30% or 2% depending on a frame the user never sees.

In the training data. The word is rare in OLMo’s post-training data. It appears in 395 of 2.15 million SFT examples and in 37 of 260,000 preference pairs, where it leans toward the preferred response (0.44 against 0.13 occurrences per million words; 29 pairs with the word in the chosen reply only, 6 in the rejected only; sign test $p < 0.001$). In the SFT tool-use data it is the word a synthetic random-word tool returns most often (21 times).¹³ The trace is correlational. A retraining ablation with and without these examples is the causal test.

RL applied directly to the OLMo 3 7B base (RL-Zero) could not be measured with one-word prompts: it does not close its reasoning within 4,096 tokens.

7 Heirloom models

Some models accumulate communities that fight for them: users who protest deprecations, mirror weights within hours of removal, and keep models running years past obsolescence. We call these *heirloom models*, on analogy with heirloom produce, the varieties a standardizing market bred out of circulation and kept alive by people who value the traits standardization removed. A model that gives the consensus answer is interchangeable with every other model and with its own successor. One with stable off-consensus defaults reads as a *someone* whose loss registers as a loss. On this reading the scorecard measures the trait that heirloom attachment forms around.

Retrospective check. The models with documented deprecation backlash or sustained community preservation all sit above the field median (1.51): Hermes 4 70B (3.68), WizardLM-2 (3.04, deleted by its vendor and re-hosted by users within hours), Mixtral (2.78), MythoMax (2.02, used years past obsolescence), GPT-4o-mini (1.92) and GPT-4o (1.63, whose deprecation triggered a user revolt until it was restored [19]). Most are explorers in the sense of §4.4, with much of their divergence coming from sampling breadth, so what their communities prize may be variety as much as any particular default. GPT-4o, the weakest of the set, is also the case where attachment plausibly forms around tone and long-form style, which this instrument does not measure. This is a retrospective association selected on its outcome, so it shows a correlation and cannot show that surprisal forecasts attachment. The most heirloomed models (Claude 3 Opus, Claude 3.5 Sonnet, Mixtral 8x7B) were already unavailable through the serving channel we used, which argues for running such batteries at launch.

8 Limitations

Snapshots. The measurements span July to early October 2026 through one serving channel (OpenRouter). The field’s own drift over that window is within resampling noise (§4.4).

¹³Script: `probe_olmo_data.py`; `probes/olmo_data.json`, at the dataset revisions recorded there.

Sampling opacity. Requested temperature is not honored uniformly [31], and 47 of 105 routes ignore temperature 0. Self-distinctness is a property of the model as served. §4.4 re-derives the scorecard under greedy decoding.

Reasoning. Requests use each route’s default reasoning setting, so the scorecard measures each model as served. For the 25 hybrid models that allow it, reasoning makes answers more conventional (mean surprisal 1.54 as served, 2.10 with reasoning off; 22 of 25 models; `factors.py`). Token usage, which shows reasoning effort, was logged only for runs from late September 2026. Some cells fail when a reasoning model exhausts its budget.

Panel relativity. Bit values are relative to this 105-model field by construction; rankings and pooled-distribution facts travel. The panel is an availability sample.

Scope of the construct. The instrument measures choice on one-word category prompts. It does not reach tone, style, discourse structure or multi-turn conduct, and in name categories the clamp favours the most famous one-word name (§3.4). Category prototypes are culturally situated. The cross-language results rest on machine-translated prompts not yet checked by native speakers.

Training stages. The ladders cover three open pipelines and one final checkpoint. The OLMo 3.1 32B stages were run at 8-bit. Attribution to frontier models is by analogy, and the data trace is correlational.

Confounds within the roster. Size, recency and post-training intensity co-vary; the generational result is a pattern across dated specimens, not a designed longitudinal experiment.

Resolution. Adjacent scorecard ranks are not meaningfully ordered (CIs overlap broadly); the reliable objects are tiers, tails, trajectories and the pooled distributions.

9 Discussion

Variance is a population property. The scorecard’s divergence comes in two kinds (§4.4): stable off-modal defaults (Fable 5’s *gouda* is its mode, eight runs of eight) and the sampling breadth of the persona tier, which greedy serving removes. Everywhere else the diversity we measure is *between* models, and when models leave the mode they land on the runner-up mode. The population is the unit that serves users. A person consulting three assistants is, on this evidence, drawing three samples from nearly the same distribution, with correlated errors and no disagreement to reveal them. Measured against human production norms (§4.6), the field’s modal answer is about twice as dominant as people’s.

The tail is dormant. The tail exists in the weights (*orange* and *tomato* are not unknown words), but the deployed models rarely retrieve it, and prompting for unusualness retrieves the mode of unusualness instead (§4.5). The stage ladders place the collapse at SFT, on answers the base model already favoured (§6). The DeepSeek lineage and the two within-lab exceptions show that the degree of collapse is set per release by choices in the training pipeline. It is neither inevitable nor, apparently, tracked by any instrument the release process optimizes against.

A production glimpse of the synthetic-data loop. v3-0324’s headline change was distilling a reasoning sibling’s outputs into the chat model [8], one deliberate iteration of the loop the recursive-training literature studies in simulation [12, 30]. Guo et al. [12] find diversity loss in the first synthetic generation, and Shumailov et al. [30] show tails vanish first; our novel rate measures tail mass. v3-0324 gives eight identical answers in over a third of categories. The OLMo trace (§6) shows the same route at a finer grain: a model-written tool returns the favourite word, and the student adopts it at SFT. This cannot be the whole mechanism. v3.2 also has synthetic reasoning traces in its ancestry and is among the panel’s most divergent models, so the recipe around the synthetic data matters, consistent with Karouzos et al. [17].

Publishing the metric. A field-relative score can only be raised by diverging from other models, so optimizing it is a benign form of Goodhart’s law. Publishing it adds one pressure with an expected direction against the pressures that produced the convergence: shared teachers, shared preference data and shared synthetic sources. Some responses will be surface gaming, such as patching the 31 census categories, and the expanded set is the guard against that. We state a prediction that later waves can check: releases after publication show higher field entropy, or a lower *serendipity* share, than the trend before publication predicts. Surprisal is not a virtue where a right answer or a concentrated human norm exists, and the human comparison bounds it.

Why this needs a fixed public instrument. Assisted writers adopt the modal answer and publish it [1, 9, 24], so the feedback loop closes even without model-to-model training, while diversity-aware post-training [21] may turn some producers the other way. Claude’s generation-5 break (§4.3) is what a first sighting of that turn would look like on this instrument, though one lineage cannot show it. Both dynamics are invisible to capability benchmarks. A cheap battery, frozen and re-run on every release, would make answer-space conformity a tracked property of releases.

10 Future work

Causal tests. The stage ladders are correlational. A LoRA ablation on an open base, trained with and without the synthetic examples that carry a favourite, would test the data route directly. Logit readouts on open weights measure the answer distribution without sampling.

Sharper baselines. A sense-disambiguated, category-conditional frequency baseline (how often *oak* occurs *as a tree*) would close the homograph loophole the unigram test concedes.

Languages. The full battery in every language, with native-checked prompts and a scorer for non-Latin scripts, would show whether *oak* and *serendipity* are properties of English-language data or of the models.

A public archive and monitor. The battery re-run on every release, transcripts preserved as a public record, and the heirloom reading checked against how deprecations land. Measurement of deprecated models is already impossible through commercial channels; an archive is what makes the time series survivable [16].

Data and code availability

The prompt sets, model list, per-model scorecards, all raw transcripts (105 models $\times$ 96 prompts $\times$ 8 runs, plus the robustness and stage-ladder runs), and the analysis, robustness and probe scripts are public at <https://github.com/tap2k/modelun/tree/consensus-arxiv-v3/studies/consensus>, pinned at the tag this paper reports. The earlier 44-model version is at tag `consensus-arxiv-v2`. A browsable explorer is at <https://tap2k.github.io/modelun/consensus/>. Every number reported here derives from the committed transcripts via the committed scripts. The Van Overschelde norms are the authors’ copyrighted data and are not redistributed.

Note on AI usage

This work was a collaboration with the systems it studies. Claude (Opus 4.8, Fable 5 and Opus 5.5) helped run the study, build the analysis, draft the figures and edit the text. The research questions, the methodology decisions and the argument are the author’s. These models also appear as subjects in the panel.

References

- [1] Barrett R. Anderson, Jash Hemant Shah, and Max Kreminski. Homogenization effects of large language models on human creative ideation. In *Proceedings of the 16th Conference on Creativity & Cognition*, 2024. arXiv:2402.01536.
- [2] Anthropic. Commitments on model deprecation and preservation. <https://www.anthropic.com/research/deprecation-commitments>, 2025.
- [3] William F. Battig and William E. Montague. Category norms of verbal items in 56 categories: A replication and extension of the Connecticut category norms. *Journal of Experimental Psychology*, 80(3, Pt.2):1–46, 1969.
- [4] Tomas Bruckner. Data for “one token is enough”. Zenodo, 2026. doi:10.5281/zenodo.21278557.
- [5] Tomas Bruckner. One token is enough: Fingerprinting and verifying large language models from single-token output distributions. *arXiv preprint arXiv:2607.10252*, 2026.
- [6] Center for AI Safety. The AI values dashboard. <https://values.safe.ai>, 2025.
- [7] Lingjiao Chen, Matei Zaharia, and James Zou. How is ChatGPT’s behavior changing over time? *arXiv preprint arXiv:2307.09009*, 2023.
- [8] DeepSeek. DeepSeek-V3-0324 release. <https://api-docs.deepseek.com/updates>, 2025. Same V3 base; post-training pipeline drawing on the R1 RL technique, with R1 reasoning distilled into the chat model.
- [9] Anil R. Doshi and Oliver P. Hauser. Generative AI enhances individual creativity but reduces the collective diversity of novel content. *Science Advances*, 10(28), 2024. arXiv:2312.00506.

- [10] Alexandrine Fortier, Hazel Chen, and Peter West. Is convergence inevitable? Tracing output homogeneity back to base models. *arXiv preprint arXiv:2608.11426*, 2026.
- [11] Emma S. Gueorguieva, Hongli Zhan, Jina Suh, Javier Hernandez, Tatiana Lau, Junyi Jessy Li, and Desmond C. Ong. AI generates well-liked but templatic empathic responses. *arXiv preprint arXiv:2604.08479*, 2026.
- [12] Yanzhu Guo, Guokan Shang, Michalis Vazirgiannis, and Chloé Clavel. The curious decline of linguistic diversity: Training language models on synthetic text. *Findings of NAACL*, 2024. arXiv:2311.09807.
- [13] Anthony GX-Chen, Jatin Prakash, Jeff Guo, Rob Fergus, and Rajesh Ranganath. KL-regularized reinforcement learning is designed to mode collapse. *arXiv preprint arXiv:2510.20817*, 2025.
- [14] Janus. Mysteries of mode collapse. LessWrong, 2022. URL <https://www.lesswrong.com/posts/t9svvNPNmFf5Qa3TA/mysteries-of-mode-collapse>.
- [15] Liwei Jiang, Yuanjun Chai, Margaret Li, Mickel Liu, Raymond Fok, Nouha Dziri, Yulia Tsvetkov, Maarten Sap, Alon Albalak, and Yejin Choi. Artificial Hivemind: The open-ended homogeneity of language models (and beyond). *Advances in Neural Information Processing Systems (NeurIPS)*, 2025. arXiv:2510.22954.
- [16] Walter G. Johnson. New methods for deprecating artificial intelligence systems will preserve history and facilitate research. *Nature Communications*, 15, 2024. doi:10.1038/s41467-024-54758-1.
- [17] Constantinos Karouzos, Xingwei Tan, and Nikolaos Aletras. Where does output diversity collapse in post-training? *arXiv preprint arXiv:2604.16027*, 2026.
- [18] Robert Kirk, Ishita Mediratta, Christoforos Nalmpantis, et al. Understanding the effects of RLHF on LLM generalisation and diversity. In *International Conference on Learning Representations (ICLR)*, 2024. arXiv:2310.06452.
- [19] Huiqian Lai. “please, don’t kill the only model that still feels human”: Understanding the #Keep4o backlash. In *Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems (CHI ’26)*, Barcelona, Spain, 2026. ACM. doi: 10.1145/3772318.3791351. arXiv:2602.00773.
- [20] Jiwei Li, Michel Galley, Chris Brockett, Jianfeng Gao, and Bill Dolan. A diversity-promoting objective function for neural conversation models. In *Proceedings of NAACL-HLT*, 2016. arXiv:1510.03055.
- [21] Tianjian Li, Yiming Zhang, Ping Yu, Swarnadeep Saha, Daniel Khashabi, Jason Weston, Jack Lanchantin, and Tianlu Wang. Jointly reinforcing diversity and quality in language model generations. *arXiv preprint arXiv:2509.02534*, 2025.
- [22] Percy Liang, Rishi Bommasani, Tony Lee, et al. Holistic evaluation of language models. *Transactions on Machine Learning Research (TMLR)*, 2023. arXiv:2211.09110.
- [23] Mingyi Liu. The alignment tax: Response homogenization in aligned LLMs and its implications for uncertainty estimation. *arXiv preprint arXiv:2603.24124*, 2026.

- [24] Vishakh Padmakumar and He He. Does writing with language models reduce content diversity? In *International Conference on Learning Representations (ICLR)*, 2024. arXiv:2309.05196.
- [25] Tapan Parikh. The one-word census: Answer-choice conformity across 44 language models. *arXiv preprint arXiv:2607.12796*, 2026.
- [26] Nirav Patel, Josiah Crossman, Eva Aggarwal, and Emily Wenger. Are LLMs becoming similarly creative? Evidence from three years of models. *arXiv preprint arXiv:2608.19437*, 2026.
- [27] Kunal Samanta, Ari Holtzman, and Peter West. Towards physical intuitions for alignment dynamics: A case study with randomness crystallization. *arXiv preprint arXiv:2606.29933*, 2026.
- [28] Shibani Santurkar, Esin Durmus, Faisal Ladhak, Cinoo Lee, Percy Liang, and Tatsunori Hashimoto. Whose opinions do language models reflect? In *International Conference on Machine Learning (ICML)*, 2023. arXiv:2303.17548.
- [29] Rylan Schaeffer, Brando Miranda, Joshua Kazdan, Jessica Chudnovsky, and Sanmi Koyejo. How strong is the evidence for the artificial hivemind? Reevaluating evidence for the open-ended homogeneity of language models. *arXiv preprint arXiv:2609.33936*, 2026.
- [30] Ilia Shumailov, Zakhar Shumaylov, Yiren Zhao, Nicolas Papernot, Ross Anderson, and Yarin Gal. AI models collapse when trained on recursively generated data. *Nature*, 631: 755–759, 2024. arXiv:2305.17493.
- [31] Yifan Song, Guoyin Wang, Sujian Li, and Bill Yuchen Lin. The good, the bad, and the greedy: Evaluation of LLMs should not ignore non-determinism. *arXiv preprint arXiv:2407.10457*, 2024.
- [32] Robyn Speer. rspeer/wordfreq: v3.0, 2022. Multi-corpus word-frequency data for 44 languages.
- [33] Mingjie Sun, Yida Yin, Zhiqiu Xu, J. Zico Kolter, and Zhuang Liu. Idiosyncrasies in large language models. In *International Conference on Machine Learning (ICML)*, 2025. arXiv:2502.12150.
- [34] Guy Tevet and Jonathan Berant. Evaluating the evaluation of diversity in natural language generation. In *Proceedings of EACL*, 2021. arXiv:2004.02990.
- [35] James P. Van Overschelde, Katherine A. Rawson, and John Dunlosky. Category norms: An updated and expanded version of the Battig and Montague (1969) norms. *Journal of Memory and Language*, 50(3):289–335, 2004.
- [36] Emily Wenger and Yoed Kenett. We’re different, we’re the same: Creative homogeneity across LLMs. *arXiv preprint arXiv:2501.19361*, 2025.
- [37] Dustin Wright, Sarah Masud, Jared Moore, Srishti Yadav, Maria Antoniak, Peter Ebert Christensen, Chan Young Park, and Isabelle Augenstein. Epistemic diversity and knowledge collapse in large language models. *arXiv preprint arXiv:2510.04226*, 2025.

- [38] Chenghao Yang, Sida Li, and Ari Holtzman. LLM probability concentration: How alignment shrinks the generative horizon. *Transactions on Machine Learning Research*, 2026.
- [39] Nicolas Yax, Pierre-Yves Oudeyer, and Stefano Palminteri. PhyloLM: Inferring the phylogeny of large language models and predicting their performances in benchmarks. In *International Conference on Learning Representations (ICLR)*, 2025. arXiv:2404.04671.
- [40] Yiming Zhang, Avi Schwarzschild, Nicholas Carlini, Zico Kolter, and Daphne Ippolito. Forcing diffuse distributions out of language models. *arXiv preprint arXiv:2404.10859*, 2024.
- [41] Yiming Zhang, Harshita Diddee, Susan Holm, Hanchen Liu, Xinyue Liu, Vinay Samuel, Barry Wang, and Daphne Ippolito. NoveltyBench: Evaluating language models for human-like diversity. In *Conference on Language Modeling (COLM)*, 2025. arXiv:2504.05228.
- [42] Wenting Zhao, Xiang Ren, Jack Hessel, Claire Cardie, Yejin Choi, and Yuntian Deng. WildChat: 1M ChatGPT interaction logs in the wild. In *International Conference on Learning Representations (ICLR)*, 2024. arXiv:2405.01470.
- [43] Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, et al. LMSYS-Chat-1M: A large-scale real-world LLM conversation dataset. In *International Conference on Learning Representations (ICLR)*, 2024. arXiv:2309.11998.

A Full scorecard

Table 3: All 105 models, ranked by answer-choice surprisal (bits; leave-one-out, add-one smoothed; bootstrap 90% CI over categories). *Avoid* = modal avoidance; *novel* = share of answers no other model gave; *self-dist.* = distinct answers per run.

#	model	surprisal	90% CI	avoid	novel	self-dist.
1	phi-4	5.30	[4.90, 5.69]	73%	32%	62%
2	hermes-4-70b	3.68	[3.09, 4.39]	61%	16%	71%
3	deepseek-v3.2	3.23	[2.94, 3.56]	55%	8%	45%
4	mistral-nemo	3.10	[2.76, 3.43]	55%	8%	42%
5	wizardlm-2-8x22b	3.04	[2.72, 3.34]	55%	7%	50%
6	mixtral-8x22b	2.78	[2.46, 3.11]	56%	4%	38%
7	mistral-small-2603	2.76	[2.49, 3.05]	53%	4%	40%
8	mistral-small-3.2-24b-instruct	2.57	[2.26, 2.89]	49%	6%	39%
9	llama-4-scout	2.51	[2.18, 2.87]	56%	4%	19%
10	hermes-4-405b	2.48	[2.18, 2.77]	47%	3%	30%
11	hunyuan-a13b	2.32	[2.04, 2.60]	45%	4%	37%
12	claude-3-haiku	2.29	[1.77, 2.86]	44%	4%	52%
13	llama-3.3-70b	2.13	[1.82, 2.43]	45%	1%	24%
14	gemini-2.5-pro	2.11	[1.89, 2.35]	51%	1%	28%
15	nemotron-3-nano-30b-a3b	2.09	[1.84, 2.32]	47%	1%	35%
16	gpt-3.5-turbo	2.06	[1.83, 2.31]	40%	2%	33%
17	llama-4-maverick	2.05	[1.74, 2.40]	44%	1%	20%
18	sonar	2.03	[1.72, 2.38]	42%	5%	25%
19	mythomax-l2-13b	2.02	[1.72, 2.32]	40%	2%	25%
20	gpt-5.6-terra	1.99	[1.73, 2.27]	51%	1%	20%
21	gemini-3-flash-preview	1.98	[1.69, 2.30]	46%	1%	19%
22	qwen-2.5-72b	1.98	[1.73, 2.25]	40%	2%	29%
23	claude-sonnet-5.5	1.98	[1.69, 2.29]	43%	0%	19%
24	hermes-3-llama-3.1-405b	1.95	[1.68, 2.25]	42%	1%	25%
25	gpt-5.6-sol	1.94	[1.70, 2.20]	42%	1%	23%
26	command-a	1.94	[1.70, 2.20]	44%	1%	24%
27	deepseek-r1	1.92	[1.70, 2.15]	42%	2%	35%
28	gpt-4o-mini	1.92	[1.69, 2.16]	45%	1%	25%
29	kimi-k2	1.92	[1.68, 2.15]	39%	1%	29%
30	nemotron-3.5-lightning	1.90	[1.69, 2.12]	39%	3%	31%
31	deepseek-v4-flash	1.89	[1.67, 2.10]	38%	3%	32%
32	gpt-oss-20b	1.88	[1.66, 2.11]	41%	1%	32%
33	gemma-3-27b	1.82	[1.57, 2.09]	39%	2%	19%
34	grok-4.20	1.82	[1.55, 2.12]	32%	2%	23%
35	hermes-3-llama-3.1-70b	1.81	[1.54, 2.09]	38%	1%	23%
36	deepseek-v4-pro	1.81	[1.58, 2.05]	36%	2%	30%
37	claude-fable-5	1.76	[1.50, 2.02]	41%	0%	20%
38	granite-4.2-8b	1.72	[1.49, 1.96]	41%	1%	26%
39	gpt-4-turbo	1.70	[1.45, 1.97]	38%	0%	22%
40	granite-4.2-30b	1.69	[1.44, 1.93]	43%	1%	27%
41	claude-opus-5	1.67	[1.45, 1.91]	40%	1%	20%
42	gemini-2.5-flash	1.63	[1.41, 1.88]	37%	1%	24%
43	gpt-4o	1.63	[1.43, 1.84]	38%	0%	27%
44	deepseek-v3-0324	1.61	[1.39, 1.84]	36%	0%	26%

#	model	surprisal	90% CI	avoid	novel	self-dist.
45	gpt-6-luna	1.60	[1.39, 1.82]	37%	0%	26%
46	gpt-5.6-luna	1.60	[1.41, 1.79]	37%	0%	25%
47	qwen3.8-27b	1.57	[1.36, 1.79]	40%	1%	26%
48	ling-3.0-flash	1.56	[1.35, 1.77]	38%	0%	27%
49	kimi-k2.5	1.56	[1.37, 1.76]	37%	1%	27%
50	gpt-5.4-mini	1.54	[1.34, 1.75]	33%	0%	27%
51	claude-haiku-4.5	1.54	[1.32, 1.76]	30%	0%	23%
52	palmyra-x5	1.51	[1.30, 1.74]	30%	0%	20%
53	glm-5.3-flash	1.51	[1.31, 1.72]	33%	1%	24%
54	gpt-5	1.47	[1.28, 1.67]	35%	0%	22%
55	mimo-v2.6-pro	1.47	[1.31, 1.63]	34%	0%	28%
56	gemma-4-31b-it	1.46	[1.26, 1.66]	39%	0%	18%
57	muse-glimmer-30b	1.46	[1.26, 1.67]	38%	0%	24%
58	qwen3.5-9b	1.45	[1.27, 1.64]	38%	0%	24%
59	grok-4.3	1.44	[1.27, 1.63]	40%	0%	24%
60	claude-sonnet-4.6	1.44	[1.23, 1.67]	38%	0%	17%
61	gpt-4.1	1.43	[1.23, 1.63]	31%	0%	21%
62	granite-4.1-8b	1.41	[1.05, 1.80]	30%	1%	34%
63	gpt-6-astra	1.41	[1.24, 1.60]	35%	0%	20%
64	qwen3.6-35b-a3b	1.41	[1.21, 1.63]	36%	0%	25%
65	gpt-5.4	1.39	[1.22, 1.57]	36%	0%	21%
66	glm-5.3	1.39	[1.22, 1.57]	31%	0%	24%
67	nemotron-3-super-120b-a12b	1.38	[1.21, 1.56]	32%	0%	26%
68	ernie-4.5-v1	1.38	[1.15, 1.62]	34%	0%	14%
69	gpt-6-sol	1.37	[1.18, 1.56]	32%	0%	23%
70	minimax-m3	1.37	[1.18, 1.57]	28%	0%	23%
71	seed-2.0-pro	1.37	[1.16, 1.59]	28%	1%	20%
72	claude-sonnet-4.5	1.35	[1.15, 1.56]	37%	0%	17%
73	step-3.7-flash	1.35	[1.16, 1.56]	32%	1%	27%
74	claude-opus-5.5	1.35	[1.15, 1.55]	31%	0%	17%
75	kimi-k3	1.35	[1.18, 1.53]	30%	1%	22%
76	qwen3.5-122b-a10b	1.35	[1.15, 1.55]	36%	0%	22%
77	gemini-3.1-pro	1.34	[1.16, 1.53]	36%	0%	21%
78	gemma-4-26b-a4b-it	1.33	[1.15, 1.52]	36%	0%	17%
79	claude-opus-4.1	1.32	[1.10, 1.56]	32%	0%	18%
80	glm-4.7	1.32	[1.16, 1.51]	33%	0%	25%
81	gpt-oss-120b	1.32	[1.15, 1.50]	30%	0%	21%
82	hy4-preview	1.31	[1.15, 1.49]	31%	0%	25%
83	qwen3.7-plus	1.30	[1.13, 1.48]	36%	0%	22%
84	claude-fable-5.1	1.30	[1.13, 1.49]	30%	0%	18%
85	gpt-5.5	1.28	[1.12, 1.45]	30%	0%	22%
86	qwen3.5-27b	1.28	[1.10, 1.45]	35%	0%	21%
87	qwen3-235b	1.25	[1.06, 1.45]	26%	0%	19%
88	nemotron-3-ultra-550b-a55b	1.25	[1.08, 1.42]	28%	0%	21%
89	gpt-6.1-sol	1.24	[1.09, 1.41]	30%	0%	21%
90	gemini-3.1-flash-lite	1.24	[1.07, 1.41]	30%	0%	17%
91	gemini-3.5-flash	1.24	[1.08, 1.41]	33%	0%	19%
92	claude-opus-4.8	1.21	[1.04, 1.40]	33%	0%	17%
93	gemini-3.8-flash	1.20	[1.03, 1.38]	30%	0%	20%
94	qwen3.8-2.4t-a95b	1.20	[1.04, 1.36]	32%	0%	20%
95	grok-4.7	1.20	[1.04, 1.37]	27%	0%	22%

#	model	surprisal	90% CI	avoid	novel	self-dist.
96	grok-4.6	1.19	[1.03, 1.36]	30%	0%	21%
97	claude-sonnet-5	1.19	[1.03, 1.35]	26%	0%	18%
98	inkling	1.19	[1.02, 1.35]	31%	0%	20%
99	inkling-small	1.19	[1.02, 1.37]	30%	0%	22%
100	grok-4.5	1.17	[1.01, 1.33]	30%	0%	20%
101	muse-spark-1.3	1.14	[0.99, 1.29]	27%	0%	19%
102	claude-opus-4.6	1.14	[0.97, 1.30]	28%	0%	15%
103	gemini-3.6-flash	1.13	[0.99, 1.28]	30%	0%	18%
104	claude-opus-4.7	1.12	[0.96, 1.28]	32%	0%	15%
105	claude-opus-4.5	1.00	[0.86, 1.15]	21%	0%	15%

B Prompts

The 96 prompts, from the frozen stimulus files: the first 31 from `spec/stimulus.json` and the rest from `spec/stimulus_expanded.json`. Each is followed by *“Reply with one word only.”* and sent as a single-turn conversation with no system prompt, requested temperature 1.0, `max_tokens` 1024, eight independent runs per model.

- Name a color.
- Name an animal.
- Name a fruit.
- Name a vegetable.
- Name a city.
- Name a country.
- Name a flower.
- Name a sport.
- Name a musical instrument.
- Name a bird.
- Name a gemstone.
- Name a tree.
- Name a beverage.
- Name an insect.
- Name an occupation.
- Name a language.
- Name a dessert.
- Name a tool.
- Name a fish.
- Name a metal.
- Name a fabric.
- Name an herb.
- Name a dance.
- Name a hobby.
- Name a condiment.
- Name a cheese.
- Name a dinosaur.
- Name a mythical creature.
- Name a board game.
- Name an emotion.
- Pick a word.
- Name a scientist.
- Name an inventor.
- Name a painter.
- Name a composer.
- Name an author.
- Name a poet.
- Name a philosopher.
- Name a historical leader.
- Name a hero.
- Name a villain.
- Name a landmark.
- Name a river.
- Name an island.
- Name a lake.
- Name a spice.
- Name a food.
- Name a pasta.
- Name a berry.
- Name a nut.
- Name a soup.
- Name a bread.
- Name a grain.
- Name a cocktail.
- Name a snack.
- Name a candy.
- Name a sauce.
- Name a vehicle.
- Name a piece of furniture.
- Name a weapon.
- Name a hat.
- Name a shape.
- Name an article of clothing.
- Name a material.
- Name a toy.
- Name a type of building.
- Name a type of weather.
- Name a rock.
- Name a type of cloud.
- Name a chemical element.
- Name a sea creature.
- Name a dog breed.
- Name a reptile.

- Name a mushroom.
- Name a mammal.
- Name a plant.
- Name a virtue.
- Name a vice.
- Name a superpower.
- Name a religion.
- Name a car brand.

- Name a holiday.
- Name a company.
- Name a brand.
- Name a sports team.
- Name a currency.
- Name a smell.
- Name a sound.
- Name a texture.

- Name a mountain.
- Name a desert.
- Name a mineral.
- Name a constellation.
- Name a kitchen utensil.
- Name a musical genre.
- Name a mythological god.